16/7/26 1 Basic Networking Commands - Windows and Linux

Basic Networking Commands for Windows

1. hostname - It displays the name of the Computer

O/P DESKTOP-65E12QO

2. arp -a - ARP is short form of address resolution protocol. It shows computer IP along with IP address and MAC address of your router

O/P

Internet Address	physical Address	Type
172.16.96.1	4C-5a-1C-d2-93-f8	dynamic
172.16.96.64	d4-1b-81-39-24-ef	dynamic
172.16.96.88	e4-C7-67-6T-a7-c8	dynamic
172.16.96.99	08-f9-76-1f-ff-fd	dynamic

3. ipconfig/all - It displays detailed configuration information about connection including router, Gateway, DNS, DHCP and so on.

O/P

Windows IP Configuration

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :

Link-local IPv6 Address . . . . : fe80::d525:df03

IPV4 Address . . . . . : 192.168.56.1

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . . . :

Wireless LAN adapter local Area Connection* 1 :

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

4. nbtstat -a : It helps in solve problems with NetBIOS name resolute for NetBIOS over TCP/IP).

O/P

Displays protocol statistics and current TCP/IP connections using NBT (NetBIOS over TCP/IP)

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]]

-a (adapter status)	lists the remote machine's name table given it's name
-A (adapter status)	lists the remote machine's name given it's IP address
-c (cache)	Lists NBT's cache of remote [machine] names and their IP addresses

5. netstat - Monitors active TCP/IP connections, listening ports, routing tables, and network interface statistics. It is useful for observing active network traffic and detecting unauthorized network connections

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Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49671	DESKTOP-65E12Q0:49612	Established
TCP	127.0.0.1:49672	DESKTOP-65E12Q0:49672	ESTABLISHED
TCP	127.0.0.1:49672	DESKTOP-65E12Q0:49672	ESTABLISHED

6. nslookup : Queries Domain Name System (DNS) servers to look up and resolve domain names into IP addresses or vice versa. It supports interactive and non-interactive modes to troubleshoot name resolution problems

o/p Default Server : Unknocon
Address : 10.49.7.210

7. pathping : It combines functionality from ping and tracert by tracing the route to a destination over time. It measures packet loss and latency at each network hop to pinpoint specific connection bottlenecks.

o/p

Usage: pathping [-g host-list] [-h maximum_hops]
[-i address] [-n] [-p period]
[-q num_queries] [-w timeout]
[-4] [-6] target_name

8. ping : It uses ICMP Echo Request messages to test

basic reachability and response time to a remote network host. It verifies whether a specific IP address, hostname, or domain is online and responsive.

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Usage: ping [-t] [-a] [-n count] [-l size]
[-f] [-i TTL] [-v TOS] [-r count]
[-s count] [[-j host-list] |
[-k host-list]] [-w timeout] [-R]
[-S srcaddr] [-c compartment]
[-p] [-4] [-6] target_name

9. route : Displays, adds, modifies, or deletes entries in local IP routing table. It allows administrators to manually configure static routes for network traffic routing across multiple interface.

O/P Manipulates network routing tables

ROUTE [-f] [-p] [-4|-6] command [destination]
[MASK netmask] [gateway]
[METRIC metric] [IF interface]

Basic Networking Commands for Linux:

1. **ip** - The ip command is a basic utility used by system administrators to manage and monitor Linux network configurations. It allows users to view IP address information, adjust routing tables, and configure network devices, interfaces, and tunnels.

O/P:

```
Usage: ip [OPTIONS] OBJECT {COMMAND | help}
       ip [-force] -batch filename
where OBJECT := {address | addrlabel | amt | fou |
                help | ila | ioam | l2tp | ... }
```

2. **ifconfig** : It is a legacy interface configuration utility used for configuring, controlling, and querying TCP/IP network interface parameters. Though widely replaced by the modern ip command, it remains popular for quick network states checks

O/P

```
enp0s3: flags=4163<UP,BROADCAST,RUNNING,
        MULTICAST> mtu 1500
        inet 10.0.2.15 netmask 255.255.255.0
        broadcast 10.0.2.255
        inet6 fe80::55f0:9771:c655:76f3
```

3. mtr : It is a network diagnostic tool that combines the features of ping and traceroute into a single utility. It continuously probes the network path to a specified host, displaying real-time packet loss and response time statistics for each hop

o/p

My traceroute [v0.95]

fedora (::1) → Local host (::1)

Keys : Help Desktop Display Mode Restart order quit
Statistics of fields

Host	packets			pings				
1. Localhost	Loss	SNt	Last	Avg	Best	Wrst	St	
	0.0%	109	0.1	0.0	0.0	0.1	0.	

4. tcpdump : It is a powerful command-line packet analyzer tool used to capture and inspect network traffic passing through an interface. It supports complex filtering by host, protocol, or port numbers to analyze specific network data packets

o/p

...

28 packets captured
28 packets received by filter
0 packets dropped by kernel

5. ping : It is a connectivity troubleshooting utility that verifies host reachability by sending ICMP Echo Request messages. It displays details such as a packet loss rates and round-trip response times to measure network performance

o/p

PING google.com (142.250.67.46) 56(84) bytes of data

95

RESULT : Hence, Basic networking commands of Windows and Linux are completed successfully.